1 Three stochastic epidemic models

We consider three discrete-time, branching-process models for the spread of an infectious disease. Time is measured in days. Let $I_t \geq 0$ denote the number of new infections (incidence) on day t, and let $R_t > 0$ be the effective reproduction number on day t.

The *serial interval weights* $w_s > 0$ $(s = 1, 2, \ldots)$ satisfy $\sum_{s=1}^{\infty} w_s = 1$ and encode the probability that a secondary infection appears s days after the primary case. Write

$$F_r = \sum_{s=1}^{r} w_s$$

for the cumulative serial interval distribution at lag r, so $1 - F_r = \sum_{s=r+1}^{\infty} w_s$ is the probability that transmission occurs after day r.

The first two models share a *dispersion parameter* $k > 0$ controlling individual-level variation in infectiousness. Small k corresponds to high heterogeneity (superspreading); as $k \to \infty$ both models collapse onto a common Poisson offspring distribution with no dispersion parameter — our third model.

Remark. Throughout, $\mathrm{NB}(\mathrm{mean} = \mu, \mathrm{disp} = k)$ denotes the negative-binomial distribution with probability mass function

$$\mathbb{P}(X = n) = \binom{n+k-1}{n} \left(\frac{k}{k+\mu}\right)^k \left(\frac{\mu}{k+\mu}\right)^n, \quad n = 0, 1, 2, \ldots,$$

mean μ, variance $\mu + \mu^2/k$, and probability generating function (pgf) $G(s) = \big(k \,/\, (k + \mu(1-s))\big)^k$. Similarly, $\mathrm{Gamma}(\alpha, \beta)$ denotes the Gamma distribution with shape α and rate β, so density $f(x) = \beta^\alpha x^{\alpha-1} e^{-\beta x}/\Gamma(\alpha)$, mean α/β, and variance α/β^2.

1.1 SSE model (Superspreading Events)

In the *superspreading-events* (SSE) model, the force of infection on day t is the weighted sum of past incidence,

$$\lambda_t = \sum_{s=1}^{\infty} w_s \, I_{t-s}, \tag{1}$$

and new infections are negative-binomially distributed:

$$I_t \mid \lambda_t \;\sim\; \mathrm{NB}(\mathrm{mean} = R_t \lambda_t,\ \mathrm{disp} = k\lambda_t)\,. \tag{2}$$

The conditional mean of I_t is $R_t\lambda_t$ and its variance is $R_t\lambda_t + R_t^2\lambda_t/k$, so the NB inflates the Poisson variance by the factor $1 + R_t\lambda_t/k$, capturing superspreading at the population level.

1.2 SSI model (Superspreading Individuals)

In the *superspreading individuals* (SSI) model, heterogeneity enters at the individual level. Each infected individual has a latent infectivity that is Gamma distributed with mean 1 and variance $1/k$.

Each cohort of I_t infected individuals then has an aggregate infectivity

$$Y_t \mid I_t \;\sim\; \mathrm{Gamma}(\mathrm{shape} = kI_t,\ \mathrm{rate} = k)\,, \tag{3}$$

so $\mathbb{E}[Y_t \mid I_t] = I_t$ and $\mathrm{Var}(Y_t \mid I_t) = I_t/k$. New infections arise as a Poisson process driven by past infectivity,

$$I_t \mid (Y_s)_{s<t} \;\sim\; \mathrm{Poisson}\left(R_t \sum_{s=1}^{\infty} w_s \, Y_{t-s}\right). \tag{4}$$

Remark. Marginalising Y_t from the SSI model recovers a negative-binomial offspring distribution (by the Poisson–Gamma mixture identity; see below), so the two models share the same mean–variance structure at the population level while differing in the *mechanism* generating overdispersion.

1.3 Poisson offspring model

In the *Poisson* offspring model the force of infection on day t is the same weighted sum as in the SSE model,

$$\lambda_t = \sum_{s=1}^{\infty} w_s\, I_{t-s}, \tag{5}$$

and new infections are Poisson distributed:

$$I_t \mid \lambda_t \;\sim\; \mathrm{Poisson}(R_t\lambda_t)\,. \tag{6}$$

The conditional mean and variance of I_t are both $R_t\lambda_t$: the Poisson is the limit of the SSE-style overdispersion vanishing ($1 + R_t\lambda_t/k \to 1$ as $k \to \infty$).

Remark. The Poisson model is the common $k \to \infty$ limit of both the SSE and the SSI models. In the SSE step the negative-binomial offspring distribution $\mathrm{NB}(R_t\lambda_t, k\lambda_t)$ collapses to $\mathrm{Poisson}(R_t\lambda_t)$ as $k \to \infty$; in the SSI step the Gamma noise on the individual infectivities degenerates to $Y_t = I_t$ a.s., again leaving $\mathrm{Poisson}(R_t\lambda_t)$ for the increment. There is no dispersion parameter; the only model parameter is R_t.

2 Extinction probability

Henceforth assume the reproduction number is constant, $R_t = R_0$. We study the probability that the epidemic goes extinct, starting from a single infectious case on day 0.

By branching-process theory, the extinction probability q is the smallest non-negative fixed point of the pgf of the offspring distribution. For the SSE and SSI models the offspring distribution is $\mathrm{NB}(\text{mean} = R_0, \text{disp} = k)$, whose pgf is $G(s) = (k/(k + R_0(1 - s)))^k$; setting $q = G(q)$ gives

$$q = \left(1 + \frac{R_0}{k}(1 - q)\right)^{-k}. \tag{7}$$

For the Poisson model the offspring distribution is $\mathrm{Poisson}(R_0)$, whose pgf is $G(s) = e^{R_0(s-1)}$; setting $q = G(q)$ gives

$$q = e^{R_0(q-1)}. \tag{8}$$

In all three models, if $R_0 \le 1$ the unique solution is $q = 1$ (extinction is certain); if $R_0 > 1$ there is a unique solution $q \in (0, 1)$, so the probability of a major outbreak is $1 - q > 0$.

2.1 Extinction probability after observing incidence on days 0 to r

Suppose we observe incidence $I_0, I_1, \ldots, I_r$ on the first $r + 1$ days, with $I_0 \ge 1$. We compute the updated extinction probability q_r for the residual epidemic — the probability that the chain triggered by the observed cases eventually dies out.

A standard branching-process argument applies in all three models: each remaining offspring of the observed cases, once born, initiates an independent fresh epidemic governed by the original offspring distribution ($\mathrm{NB}(R_0,\, k)$ for SSE/SSI, $\mathrm{Poisson}(R_0)$ for Poisson) and goes extinct with the standard probability q (the smallest fixed point of G). The conditioning event constrains only the offspring of the observed cases on days $0, \ldots, r$; it does not propagate to later generations. Therefore

$$q_r = G_r(q),$$

where G_r is the pgf of the (conditional) total remaining offspring of the observed cases. The three models differ in the form of G_r.

2.1.1 SSE model

In the SSE model the offspring process decomposes into independent contributions: each of the I_s individuals infected on day s generates a number of offspring at lag ℓ that is NB(mean $= R_0 w_\ell$, disp $= k w_\ell$), independently across individuals, donor days, and lags. By the NB closure property under a common success probability $p = k w_\ell/(k w_\ell + R_0 w_\ell) = k/(k + R_0)$ (independent of ℓ), the population-level $\mathrm{NB}(R_0\lambda_t, k\lambda_t)$ law of I_t is recovered by summing all per-individual-per-lag contributions.

Conditioning on the observed counts $I_0, \ldots, I_r$ pins down only those contributions (s, ℓ) with $s + \ell \leq r$. Per-individual-per-lag contributions with $s \in \{0, \ldots, r\}$ and $\ell > r - s$ are independent of the observations and retain their original distributions. The total remaining offspring of the observed cases is therefore

$$\sum_{s=0}^{r}\sum_{i=1}^{I_s}\sum_{\ell=r-s+1}^{\infty} \mathrm{NB}(\text{mean} = R_0 w_\ell,\ \text{disp} = k w_\ell) = \mathrm{NB}(\text{mean} = R_0\Lambda,\ \text{disp} = k\Lambda), \tag{9}$$

where the pooled effective weight is

$$\Lambda \;=\; \sum_{s=0}^{r} I_s\,(1 - F_{r-s}). \tag{10}$$

With the convention $F_0 = 0$, the $s = r$ term enters with full weight, since none of its offspring have had a chance to appear yet.

The pgf is $G_r(s) = \big(1 + (R_0/k)(1 - s)\big)^{-k\Lambda} = G(s)^\Lambda$, so

$$q_r \;=\; G_r(q) \;=\; q^\Lambda \;=\; q^{\sum_{s=0}^{r} I_s\,(1-F_{r-s})}. \tag{11}$$

The earlier special case (a single case on day 0 followed by r days without cases) corresponds to $I_0 = 1$, $I_1 = \cdots = I_r = 0$, giving $\Lambda = 1 - F_r$ and recovering $q_r = q^{1-F_r}$. In the same direction, as the zero streak extends ($r \to \infty$) the contributions $I_s\,(1 - F_{r-s})$ all vanish and $q_r \to 1$: extinction approaches certainty.

2.1.2 SSI model

For the SSI model the joint posterior of $(Y_0, \ldots, Y_{r-1})$ is in general not available in closed form: as soon as $I_d > 0$ on at least one day $d \in \{1, \ldots, r\}$, the day-d rate $R_0 \sum_{s=1}^{d-1} w_{d-s} Y_s$ couples the latents. We therefore present an MCMC estimator of q_r here as the canonical SSI workflow. For the special cases of cases on day 0 only, on day 0 and one later day, and on day 0 and two later days, closed forms are however available; these are deferred to § 4 below.

MCMC posterior. We sample $(Y_0, \ldots, Y_{r-1})$ from their joint posterior given $(I_0, \ldots, I_r)$ using a Hamiltonian Monte Carlo sampler (implemented in PyMC). Note that Y_r, the aggregate infectivity of the day-r cohort, has not yet contributed to any observed outcome; its posterior equals its prior,

$$Y_r \;\sim\; \mathrm{Gamma}(kI_r,\, k),$$

and is sampled independently.

Remaining offspring from observed cases. Given $(Y_s)_{s=0}^{r}$, the future infections generated by the observed cases arrive as independent Poisson streams: cohort s contributes at lag $\ell \geq r - s + 1$ at rate $R_0 w_\ell Y_s$. Summing over cohorts and lags gives total remaining offspring

$$I_{\text{rem}} \mid (Y_s)_{s=0}^{r} \;\sim\; \mathrm{Poisson}(R_0\,\Lambda)\,, \qquad \Lambda \;=\; \sum_{s=0}^{r} Y_s\,(1 - F_{r-s}), \tag{12}$$

exactly the same pooled-weight formula as in the SSE model.

Extinction probability and MCMC estimator. Each remaining offspring initiates an independent fresh epidemic that goes extinct with the standard probability q. Hence the conditional extinction probability of the residual epidemic, given the latent infectivities, is the pgf of Poisson($R_0\Lambda$) evaluated at q:

$$q_r \mid (Y_s)_{s=0}^r \;=\; e^{R_0\,\Lambda\,(q-1)}. \tag{13}$$

Marginalising over the posterior of (Y_s) and estimating by a Monte Carlo average over M MCMC draws,

$$\hat{q}_r \;=\; \frac{1}{M}\sum_{m=1}^{M} e^{R_0\,\Lambda^{(m)}\,(q-1)}, \qquad \widehat{\mathrm{PMO}} \;=\; 1-\hat{q}_r, \tag{14}$$

where $\Lambda^{(m)} = \sum_{s=0}^{r} Y_s^{(m)}(1-F_{r-s})$.

Remark (Consistency with the day-0-only analytic formula)**.** When $I_1 = \cdots = I_r = 0$ the observations force $Y_1 = \cdots = Y_r = 0$, so $\Lambda = Y_0(1-F_r)$ and the posterior is $Y_0 \sim \mathrm{Gamma}(kI_0,\, k+R_0F_r)$. Taking the expectation of $e^{R_0Y_0(1-F_r)(q-1)}$ with respect to this posterior via the Gamma moment-generating function gives

$$q_r = \left(1-\frac{R_0(1-F_r)(q-1)}{k+R_0F_r}\right)^{-kI_0} = \left(1+\frac{R_0(1-F_r)}{k+R_0F_r}(1-q)\right)^{-kI_0},$$

which matches the closed-form result (29).

2.1.3 Poisson model

The argument of § 2.1.1 carries over verbatim with NB replaced by Poisson. Each of the I_s individuals infected on day s generates a Poisson(R_0w_ℓ) number of offspring at lag ℓ, independently across individuals, donor days, and lags. By Poisson closure under summation the total remaining offspring of the observed cases is

$$\sum_{s=0}^{r}\sum_{i=1}^{I_s}\sum_{\ell=r-s+1}^{\infty} \mathrm{Poisson}(R_0w_\ell) \;=\; \mathrm{Poisson}(R_0\,\Lambda)\,, \tag{15}$$

with the same pooled effective weight $\Lambda = \sum_{s=0}^{r} I_s\,(1-F_{r-s})$ as in the SSE model. The pgf is $G_r(s) = e^{R_0\Lambda(s-1)} = G(s)^\Lambda$, so

$$q_r \;=\; G_r(q) \;=\; q^\Lambda, \tag{16}$$

identical in functional form to (11) but with q the Poisson extinction root (8). In particular, both (11) and (16) are closed-form for any observed history.

3 Probability of major outbreak under model uncertainty

So far we have computed the probability of a major outbreak conditional on a chosen model (SSE, SSI, or Poisson). In practice the underlying mechanism generating overdispersion is unknown, and we may wish to combine the candidate models via Bayesian model averaging. Let $\mathcal{M}$ denote a finite set of candidate models with prior probabilities $\{\pi_M\}_{M\in\mathcal{M}}$ satisfying $\sum_{M\in\mathcal{M}}\pi_M = 1$. Two natural choices are the two-model set $\mathcal{M} = \{\mathrm{SSE}, \mathrm{SSI}\}$ (a single k, varying only the noise mechanism) and a wider ensemble including several SSE and SSI specifications with different k values together with the Poisson limit. Given the observed history $(I_0,\ldots,I_r)$, Bayes' theorem yields posterior model probabilities

$$\pi_M^\star \;=\; \frac{\pi_M\,L_M}{\sum_{M'\in\mathcal{M}}\pi_{M'}\,L_{M'}}, \qquad L_M \;=\; \mathbb{P}_M(I_1,\ldots,I_r \mid I_0), \tag{17}$$

and the model-averaged probability of a major outbreak is the posterior expectation of the per-model PMOs,

$$\mathrm{PMO}_{\mathrm{avg}} \;=\; \sum_{M\in\mathcal{M}} \pi_M^\star\,\mathrm{PMO}_M, \tag{18}$$

where $\mathrm{PMO}_M = 1-q_r^M$ as derived above. Specialising to the two-model case $\mathcal{M} = \{\mathrm{SSE}, \mathrm{SSI}\}$, (17) and (18) become the familiar pair

$$\pi_{\mathrm{SSE}}^\star \;=\; \frac{\pi_{\mathrm{SSE}}\,L_{\mathrm{SSE}}}{\pi_{\mathrm{SSE}}\,L_{\mathrm{SSE}}+\pi_{\mathrm{SSI}}\,L_{\mathrm{SSI}}}, \qquad \mathrm{PMO}_{\mathrm{avg}} \;=\; \pi_{\mathrm{SSE}}^\star\,\mathrm{PMO}_{\mathrm{SSE}} + \pi_{\mathrm{SSI}}^\star\,\mathrm{PMO}_{\mathrm{SSI}}. \tag{19}$$

3.1 Likelihood of the observed history

SSE model. Conditional on past incidence, the day-t count satisfies $I_t \mid I_{<t} \sim \mathrm{NB}(\text{mean} = R_0\lambda_t, \text{disp} = k\lambda_t)$ with $\lambda_t = \sum_{s=1}^{t-1} w_{t-s} I_s$, so the joint likelihood factorises into a product of negative-binomial pmfs,

$$L_{\mathrm{SSE}}(I_1, \ldots, I_r \mid I_0) \;=\; \prod_{t=1}^{r} \mathrm{NB}\big(I_t;\, R_0\lambda_t,\, k\lambda_t\big). \tag{20}$$

This closed form is available for any history.

Poisson model. Likewise, the day-t count satisfies $I_t \mid I_{<t} \sim \mathrm{Poisson}(R_0\lambda_t)$, so the joint likelihood factorises into a product of Poisson pmfs,

$$L_{\mathrm{Poisson}}(I_1, \ldots, I_r \mid I_0) \;=\; \prod_{t=1}^{r} \mathrm{Poisson}\big(I_t;\, R_0\lambda_t\big), \tag{21}$$

closed-form for any history. As with (20), the convention is $\log L = -\infty$ whenever $\lambda_t = 0$ but $I_t > 0$.

SSI model, closed-form cases. Closed forms are available in the same three special cases as for q_r (cases on day 0 only, on day 0 and one later day, on day 0 and two later days); these are derived in § 4 below.

SSI model, general histories. The likelihood involves the latent infectivities $(Y_s)_{s=0}^{r-1}$ and is in general not available in closed form. Unfolding the SSI generative dependence $I_0 \to Y_0 \to I_1 \to Y_1 \to \cdots \to I_r$ as a Markov chain, the joint density of latents and observations factorises as

$$p(Y_{0:r-1},\, I_{1:r} \mid I_0) \;=\; \prod_{t=0}^{r-1} p_Y(Y_t \mid I_t) \;\cdot\; \prod_{t=1}^{r} p_I\big(I_t \mid Y_{0:t-1}\big), \tag{22}$$

where $p_Y(\,\cdot \mid I_t) = \mathrm{Gamma}(\cdot;\, kI_t,\, k)$ is the SSI generative density of Y_t given the cohort size on day t, and $p_I(\,\cdot \mid Y_{0:t-1}) = \mathrm{Poisson}(\cdot;\, R_0 \sum_{s=1}^{t} w_s Y_{t-s})$. Marginalising the latents,

$$L_{\mathrm{SSI}}(I_{1:r} \mid I_0) \;=\; \int \prod_{t=0}^{r-1} p_Y(Y_t \mid I_t) \prod_{t=1}^{r} p_I(I_t \mid Y_{0:t-1})\, \mathrm{d}Y_{0:r-1}. \tag{23}$$

Days s with $I_s = 0$ collapse Y_s to a point mass at 0, and Y_r does not enter L_{SSI} (it would only contribute to $I_{r+1}, I_{r+2}, \ldots$), so the integration runs effectively over $\mathcal{T} = \{t \in \{0, \ldots, r-1\} : I_t > 0\}$. Three MCMC-based estimators of (23) are derived in § 5.

3.2 Simulation-based estimator

For general histories, where the SSI likelihood is unavailable in closed form, we use a rejection-sampling estimator that simultaneously realises the posterior model weighting. We describe it for $\mathcal{M} = \{\mathrm{SSE}, \mathrm{SSI}\}$; it extends to any finite $\mathcal{M}$ by replacing the Binomial split below with a Multinomial split by prior, and the acceptance-rate argument carries over unchanged. Repeat the following in batches of size b:

1. Draw the per-batch model split as a Binomial: $b_{\mathrm{SSE}} \sim \mathrm{Bin}(b, \pi_{\mathrm{SSE}})$ and $b_{\mathrm{SSI}} = b - b_{\mathrm{SSE}}$.

2. Run b_{SSE} vectorised SSE forward simulations from $t = 0$ seeded with I_0 index cases, retaining only those whose simulated incidence on days $1, \ldots, r$ exactly matches the observed $I_1, \ldots, I_r$. Likewise for b_{SSI} SSI simulations.

3. For each retained simulation, continue forward until either the single-step incidence reaches the major-outbreak threshold (resolved as "major") or the most recent $L = \mathrm{len}(w)$ steps are all zero (resolved as "extinct").

Continue accumulating batches until a target number of resolved-and-matching simulations is reached. The acceptance rate of model M's simulations is L_M in expectation; with a Binomial split by prior, the expected number of accepted M-sims is proportional to $\pi_M L_M$, so the composition of the accepted pool matches the posterior $\pi^\star_M$. The pooled estimator

$$\widehat{\mathrm{PMO}}_{\mathrm{avg}} \;=\; \frac{N^{\mathrm{maj}}_{\mathrm{SSE}} + N^{\mathrm{maj}}_{\mathrm{SSI}}}{N^{\mathrm{acc}}_{\mathrm{SSE}} + N^{\mathrm{acc}}_{\mathrm{SSI}}} \;=\; \widehat{\pi}^\star_{\mathrm{SSE}}\,\widehat{\mathrm{PMO}}_{\mathrm{SSE}} + \widehat{\pi}^\star_{\mathrm{SSI}}\,\widehat{\mathrm{PMO}}_{\mathrm{SSI}} \tag{24}$$

follows immediately, where $\widehat{\pi}^\star_M = N^{\mathrm{acc}}_M/(N^{\mathrm{acc}}_{\mathrm{SSE}} + N^{\mathrm{acc}}_{\mathrm{SSI}})$ is the implied posterior estimate and $\widehat{\mathrm{PMO}}_M = N^{\mathrm{maj}}_M/N^{\mathrm{acc}}_M$ is the per-model PMO from the accepted M-sims.

4 SSI model: closed forms for histories with cases on a few days

For the SSI model, when non-zero incidence is observed on day 0 and on a small number of later days, the joint posterior of the latent infectivities still factorises (or factorises as a finite mixture) and both the model evidence L_{SSI} and the residual extinction probability q_r admit closed forms. This section treats three such cases in turn: case (i) (§ 4.1), incidence on day 0 only; case (ii) (§ 4.2), incidence on day 0 and one later day; and case (iii) (§ 4.3), incidence on day 0 and two later days.

Strategy. Each derivation follows the same five steps:

1. *Conditional likelihood given the latents.* Multiply the per-day Poisson pmfs to obtain $\mathbb{P}(\mathrm{obs} \mid Y_0, Y_{i_1}, \ldots, Y_{i_n})$.
2. *Joint density of latents and observation.* Multiply by the independent Gamma priors $Y_s \sim \mathrm{Gamma}(kI_s, k)$ to get $p(Y_0, Y_{i_1}, \ldots, Y_{i_n}, \mathrm{obs})$, with all normalising constants visible.
3. *Likelihood (model evidence).* Integrate the joint density over the latents using the Gamma normalisation $\int_0^\infty Y^{\alpha-1}e^{-\beta Y}\,\mathrm{d}Y = \Gamma(\alpha)/\beta^\alpha$ in each variable.
4. *Posterior of the latents.* Divide the joint density by the likelihood to read off the posterior; in the cases below this yields independent Gammas (or a finite mixture of independent-Gamma products in case (iii)).
5. *Residual extinction probability and PMO.* The conditional extinction probability given the latents is $q_r \mid (Y_s) = e^{-R_0(1-q)\Lambda}$ with $\Lambda = \sum_{s=0}^r Y_s(1-F_{r-s})$ (cf. the SSI MCMC derivation in § SSI model above); marginalising over the posterior via the Gamma MGF $\mathbb{E}[e^{-tX}] = (1+t/\beta)^{-\alpha}$ for $X \sim \mathrm{Gamma}(\alpha, \beta)$ gives a closed-form q_r, and the probability of a major outbreak is $\mathrm{PMO} = 1 - q_r$.

A key identity used throughout is that $I_s = 0$ on day s forces $Y_s = 0$ (since $Y_s \mid I_s \sim \mathrm{Gamma}(kI_s, k)$ collapses to a point mass at 0), so among the latents only those tied to non-zero observed days are non-degenerate. Likewise, the SSE likelihood (20) is closed-form for any history, so each subsection below ends with the model-averaged PMO via Bayes' theorem from the per-model likelihoods and PMOs.

4.1 Case (i): $I_0 \geq 1$, $I_1 = \cdots = I_r = 0$

Only Y_0 is non-degenerate; we abbreviate $y = Y_0$.

Step 1: conditional likelihood. Given y, I_d is independently $\mathrm{Poisson}(R_0 w_d y)$ for $d = 1, \ldots, r$, so

$$\mathbb{P}(I_1 = \cdots = I_r = 0 \mid y) \;=\; \prod_{d=1}^r e^{-R_0 w_d y} \;=\; e^{-R_0 F_r y}. \tag{25}$$

Step 2: joint density. Multiplying (25) by the prior $y \sim \mathrm{Gamma}(kI_0, k)$,

$$p(y, \mathrm{obs}) \;=\; \frac{k^{kI_0}}{\Gamma(kI_0)}\, y^{kI_0-1}\, e^{-(k+R_0F_r)y}. \tag{26}$$

Step 3: likelihood. Integrating y out of (26) via the Gamma normalisation,

$$L_{\mathrm{SSI}}^{(\mathrm{i})} \;=\; \frac{k^{kI_0}}{\Gamma(kI_0)} \cdot \frac{\Gamma(kI_0)}{(k+R_0F_r)^{kI_0}} \;=\; \left(1+\frac{R_0F_r}{k}\right)^{-kI_0}. \tag{27}$$

Step 4: posterior. Dividing (26) by (27) every constant cancels except the Gamma rate and shape, giving

$$y \mid \mathrm{obs} \;\sim\; \mathrm{Gamma}(kI_0,\; k+R_0F_r). \tag{28}$$

The posterior rate $k+R_0F_r > k$ reflects the fact that observing no early transmission is evidence that y is small: the index cohort is likely a low-infectivity one.

Step 5: residual extinction probability and PMO. The conditional remaining-offspring rate is $R_0\Lambda = R_0(1-F_r)y$, so the conditional extinction probability is $q_r \mid y = e^{-R_0(1-q)(1-F_r)y}$. Marginalising over the posterior (28) via the Gamma MGF,

$$q_r \;=\; \mathbb{E}\Big[e^{-R_0(1-q)(1-F_r)y}\Big] \;=\; \left(1+\frac{R_0(1-F_r)(1-q)}{k+R_0F_r}\right)^{-kI_0}, \tag{29}$$

and $\mathrm{PMO} = 1-q_r$.

Closed-form SSE counterpart. Under the SSE model, $I_t \sim \mathrm{NB}(R_0I_0w_t,\, kI_0w_t)$ independently across $t=1,\dots,r$ when $I_1=\cdots=I_r=0$, so the general likelihood (20) reduces to

$$L_{\mathrm{SSE}} \;=\; \prod_{t=1}^{r}\left(\frac{k}{k+R_0}\right)^{kI_0w_t} \;=\; \left(1+\frac{R_0}{k}\right)^{-kI_0F_r}. \tag{30}$$

The two likelihoods share the same functional shape but with different exponents: the SSE exponent kI_0F_r is reduced by $F_r \le 1$ relative to the SSI exponent kI_0, reflecting the fact that the SSE total over days $1,\dots,r$ has dispersion kI_0F_r whereas the SSI total has dispersion kI_0. Combined with the SSE PMO $1-q^{I_0(1-F_r)}$ from § SSE model and the SSI PMO (29), the model-averaged PMO is

$$\mathrm{PMO}_{\mathrm{avg}} \;=\; \pi^{\star}_{\mathrm{SSE}}\left(1-q^{I_0(1-F_r)}\right) + \pi^{\star}_{\mathrm{SSI}}\left[1-\left(1+\frac{R_0(1-F_r)(1-q)}{k+R_0F_r}\right)^{-kI_0}\right]. \tag{31}$$

When $r=0$, both likelihoods reduce to 1, the posterior reduces to the prior, and the model-averaged PMO is just the prior-weighted average of the two per-model PMOs.

Remark (Consistency with the MCMC formula)**.** When $I_1=\cdots=I_r=0$, the observations force $Y_1=\cdots=Y_r=0$, so $\Lambda = Y_0(1-F_r)$ and the general MCMC posterior of § SSI model collapses to (28). Taking the expectation of $e^{-R_0(1-q)(1-F_r)Y_0}$ with respect to that posterior via the Gamma MGF recovers (29).

4.2 Case (ii): $I_0, I_i \ge 1$ for one later day i, all others zero

We now extend to $I_0 \ge 1$, $I_i \ge 1$ for some $i \in \{1,\dots,r\}$, and $I_d = 0$ for $d \in \{1,\dots,r\}\setminus\{i\}$. Only Y_0 and Y_i are non-degenerate.

Step 1: conditional likelihood. Given (Y_0, Y_i), the day-d Poisson rate is $R_0w_dY_0$ for $d \in \{1,\dots,i\}$ and $R_0(w_dY_0 + w_{d-i}Y_i)$ for $d \in \{i+1,\dots,r\}$. Multiplying the day-d Poisson pmfs,

$$\mathbb{P}(\mathrm{obs}\mid Y_0,Y_i) = \underbrace{\prod_{d=1}^{i-1} e^{-R_0w_dY_0}}_{=\,e^{-R_0F_{i-1}Y_0}} \cdot \underbrace{\frac{(R_0w_iY_0)^{I_i}}{I_i!}\,e^{-R_0w_iY_0}}_{\text{day-}i\text{ pmf}} \cdot \underbrace{\prod_{d=i+1}^{r} e^{-R_0(w_dY_0+w_{d-i}Y_i)}}_{=\,e^{-R_0(F_r-F_i)Y_0}\,e^{-R_0F_{r-i}Y_i}}.$$

The Y_0 exponentials collapse via $F_{i-1}+w_i+(F_r-F_i) = F_r$, so

$$\mathbb{P}(\mathrm{obs}\mid Y_0,Y_i) \;=\; \frac{(R_0w_iY_0)^{I_i}}{I_i!}\,e^{-R_0F_rY_0}\,e^{-R_0F_{r-i}Y_i}. \tag{32}$$

Step 2: joint density. Multiplying (32) by the priors $Y_0 \sim \text{Gamma}(kI_0, k)$ and $Y_i \sim \text{Gamma}(kI_i, k)$ (independent),

$$\begin{aligned} p(Y_0, Y_i, \text{obs}) = & \underbrace{\frac{(R_0 w_i)^{I_i}}{I_i!} \cdot \frac{k^{kI_0+kI_i}}{\Gamma(kI_0)\,\Gamma(kI_i)}}_{\text{constants}} \\ & \times \underbrace{Y_0^{kI_0+I_i-1}\, e^{-(k+R_0F_r)Y_0}}_{\text{Gamma kernel in } Y_0} \cdot \underbrace{Y_i^{kI_i-1}\, e^{-(k+R_0F_{r-i})Y_i}}_{\text{Gamma kernel in } Y_i}. \end{aligned} \tag{33}$$

Step 3: likelihood. Integrating each Gamma kernel of (33) via the Gamma normalisation gives

$$L_{\text{SSI}}^{(\text{ii})} \;=\; \frac{(R_0 w_i)^{I_i}}{I_i!}\, \frac{\Gamma(kI_0 + I_i)}{\Gamma(kI_0)}\, \frac{k^{kI_0}}{(k+R_0F_r)^{kI_0+I_i}} \left(\frac{k}{k+R_0F_{r-i}}\right)^{kI_i}. \tag{34}$$

Setting $I_i = 0$ collapses the Γ-ratio and the second parenthesis to 1, recovering (27).

Step 4: posterior. Dividing (33) by (34), every constant cancels and the two Gamma kernels normalise independently:

$$Y_0 \mid \text{obs} \;\sim\; \text{Gamma}(kI_0 + I_i,\; k + R_0F_r), \qquad Y_i \mid \text{obs} \;\sim\; \text{Gamma}(kI_i,\; k + R_0F_{r-i}). \tag{35}$$

The shape of Y_0 is enriched by I_i: each observed offspring of the index cohort acts as a Bayesian data point tilting Y_0 toward higher values. The Y_i posterior is the day-0-only update (28) applied to the day-i cohort over its residual horizon $r - i$; setting $i = r$ gives $F_{r-i} = F_0 = 0$, so Y_r collapses to its prior, since the day-r cohort has not had a chance to contribute.

Step 5: residual extinction probability and PMO. The remaining-offspring rate is $R_0\Lambda$ with $\Lambda = Y_0(1-F_r)+Y_i(1-F_{r-i})$, so the conditional extinction probability is $q_r \mid (Y_0, Y_i) = e^{-R_0(1-q)\Lambda}$. Marginalising over the two independent Gamma factors of (35) via the Gamma MGF gives

$$q_r \;=\; \left(1 + \frac{R_0(1-F_r)(1-q)}{k+R_0F_r}\right)^{-(kI_0+I_i)} \left(1 + \frac{R_0(1-F_{r-i})(1-q)}{k+R_0F_{r-i}}\right)^{-kI_i}, \tag{36}$$

and $\text{PMO} = 1 - q_r$.

Remark (Reduction to case (i))**.** Setting $I_i = 0$ in (36) collapses the second factor to 1 and the first factor's exponent to $-kI_0$, recovering (29); likewise (34) reduces to (27). As the observation horizon extends ($r \to \infty$ with i fixed), both factors of (36) tend to 1 and $q_r \to 1$.

4.3 Case (iii): $I_0, I_i, I_j \geq 1$ for two later days $i < j$, all others zero

Finally, take $I_0 \geq 1$, $I_i \geq 1$, $I_j \geq 1$ for some $1 \leq i < j \leq r$, and $I_d = 0$ for $d \in \{1, \ldots, r\} \setminus \{i, j\}$. Now only Y_0, Y_i, Y_j are non-degenerate.

Step 1: conditional likelihood. The conditional likelihood acquires an extra factor for the day-j Poisson pmf, whose rate $R_0(w_jY_0 + w_{j-i}Y_i)$ couples Y_0 and Y_i. Repeating the telescoping argument of case (ii) for each latent separately,

$$\mathbb{P}(\text{obs} \mid Y_0, Y_i, Y_j) \;=\; \frac{(R_0 w_i Y_0)^{I_i}}{I_i!}\, \frac{\big(R_0(w_jY_0 + w_{j-i}Y_i)\big)^{I_j}}{I_j!}\, e^{-R_0F_rY_0}\, e^{-R_0F_{r-i}Y_i}\, e^{-R_0F_{r-j}Y_j}. \tag{37}$$

Expanding the day-j rate via the binomial theorem,

$$\big(w_jY_0 + w_{j-i}Y_i\big)^{I_j} \;=\; \sum_{m=0}^{I_j} \binom{I_j}{m}\, w_j^{I_j-m}\, w_{j-i}^{m}\, Y_0^{I_j-m}\, Y_i^{m}. \tag{38}$$

Remark (Latent-partition interpretation)**.** Equivalently, by Poisson superposition we may regard I_j as the sum of two independent Poissons — one with mean $R_0 w_j Y_0$ counting day-j infections that originated from the day-0 cohort, and one with mean $R_0 w_{j-i} Y_i$ counting those from the day-i cohort. Each term m in (38) is the joint probability that exactly $I_j - m$ originate from Y_0 and m from Y_i.

Step 2: joint density. Multiplying (37) by the three independent priors Gamma(kI_0, k), Gamma(kI_i, k), Gamma(kI_j, k) and inserting (38),

$$
\begin{aligned}
p(Y_0, Y_i, Y_j, \mathrm{obs}) = {} & \frac{(R_0 w_i)^{I_i}\, R_0^{I_j}}{I_i!\, I_j!} \frac{k^{kI_0 + kI_i + kI_j}}{\Gamma(kI_0)\,\Gamma(kI_i)\,\Gamma(kI_j)} Y_j^{kI_j - 1}\, e^{-(k + R_0 F_{r-j}) Y_j} \\
& \times \sum_{m=0}^{I_j} \binom{I_j}{m} w_j^{I_j - m}\, w_{j-i}^{m}\, Y_0^{kI_0 + I_i + I_j - m - 1}\, e^{-(k + R_0 F_r) Y_0}\, Y_i^{kI_i + m - 1}\, e^{-(k + R_0 F_{r-i}) Y_i}.
\end{aligned}
\tag{39}
$$

This is a constant times a Gamma kernel in Y_j (the same in every mixture component) times a finite mixture (over $m \in \{0, \ldots, I_j\}$) of Gamma-product kernels in (Y_0, Y_i).

Step 3: likelihood. Integrating (39) via the Gamma normalisation in each variable, define the per-mixture-component constants

$$
c_m \;:=\; \binom{I_j}{m} w_j^{I_j - m}\, w_{j-i}^{m}\, \frac{\Gamma(kI_0 + I_i + I_j - m)}{(k + R_0 F_r)^{kI_0 + I_i + I_j - m}}\, \frac{\Gamma(kI_i + m)}{(k + R_0 F_{r-i})^{kI_i + m}}. \tag{40}
$$

Cancelling the $\Gamma(kI_j)$ factors and grouping k^{kI_j} with the Y_j-rate gives

$$
L_{\mathrm{SSI}}^{(\mathrm{iii})} \;=\; \frac{(R_0 w_i)^{I_i}\, R_0^{I_j}}{I_i!\, I_j!}\, \frac{k^{kI_0 + kI_i}}{\Gamma(kI_0)\,\Gamma(kI_i)} \left(\frac{k}{k + R_0 F_{r-j}} \right)^{kI_j} \sum_{m=0}^{I_j} c_m. \tag{41}
$$

Setting $I_j = 0$ collapses the sum to its $m = 0$ term and the $(k/(k + R_0 F_{r-j}))^{kI_j}$ factor to 1, recovering (34).

Step 4: posterior. Dividing (39) by (41): the Y_j-Gamma rate-and-shape pair appears in both numerator and denominator and gives the Gamma$(Y_j \mid kI_j, k + R_0 F_{r-j})$ density. Within the sum, the constants outside cancel; what remains in the mth term is c_m (numerator) divided by $\sum_{m'} c_{m'}$ (denominator), times the two unnormalised (Y_0, Y_i) Gamma kernels each with their proper normalising constants supplied directly from the c_m in the denominator. The result is a finite mixture of independent Gamma products,

$$
\begin{aligned}
p(Y_0, Y_i, Y_j \mid \mathrm{obs}) \;=\; & \mathrm{Gamma}\big(Y_j \,\big|\, kI_j,\; k + R_0 F_{r-j}\big) \\
& \times \sum_{m=0}^{I_j} \rho_m\, \mathrm{Gamma}\big(Y_0 \,\big|\, kI_0 + I_i + I_j - m,\; k + R_0 F_r\big)\, \mathrm{Gamma}\big(Y_i \,\big|\, kI_i + m,\; k + R_0 F_{r-i}\big),
\end{aligned}
\tag{42}
$$

with normalised mixture weights

$$
\rho_m \;=\; \frac{c_m}{\sum_{m'=0}^{I_j} c_{m'}}. \tag{43}
$$

The mixture index m tracks how many of the I_j day-j infections are attributed to Y_i versus Y_0; the weights ρ_m encode which partition is most consistent with the prior infectivity distributions, the serial-interval weights w_j, w_{j-i}, and the dispersion k.

Step 5: residual extinction probability and PMO. The remaining-offspring rate is now $R_0 \Lambda = R_0[Y_0(1 - F_r) + Y_i(1 - F_{r-i}) + Y_j(1 - F_{r-j})]$, so the conditional extinction probability is $q_r \mid (Y_0, Y_i, Y_j) = e^{-R_0(1-q)\Lambda}$. Marginalising over the posterior (42), Y_j's expectation factors out (its density is independent of m); within each mixture component the (Y_0, Y_i)-densities are independent Gammas whose MGFs evaluate term-by-term, giving

$$
q_r \;=\; \beta_j^{-kI_j} \sum_{m=0}^{I_j} \rho_m\, \beta_0^{-(kI_0 + I_i + I_j - m)}\, \beta_i^{-(kI_i + m)}, \tag{44}
$$

where, for $a \in \{0, i, j\}$ (with the convention $F_{r-0} = F_r$),

$$\beta_a \;=\; 1 + \frac{R_0(1-q)(1-F_{r-a})}{k + R_0 F_{r-a}}.$$

Remark (Reductions to simpler cases)**.** When $I_j \to 0$ the sum collapses to its $m = 0$ term ($\rho_0 = 1$) and $\beta_j^{-kI_j} \to 1$: (44) reduces to (36). When also $I_i \to 0$, the exponent on β_i in (36) vanishes and the exponent on β_0 reduces from $-(kI_0 + I_i)$ to $-kI_0$, recovering (29). Similarly (41) reduces to (34) and then to (27).

Computational note. The c_m involve Γ factors of size up to $kI_0 + I_i + I_j$, so for moderate I_j a log-domain computation ($\log\Gamma$, logsumexp) is recommended. The number of sum terms is $I_j + 1$, typically small for the histories of interest.

5 MCMC-based estimators of the SSI marginal likelihood

We describe three estimators of the integral (23). Writing $\pi(Y) = \prod_{t\in\mathcal{T}} p_Y(Y_t \mid I_t)$ for the integrand's prior factor restricted to the active days $\mathcal{T} = \{t \in \{0, \dots, r-1\} : I_t > 0\}$ (a product of independent Gammas, treating the observed cohort sizes I_t as fixed shape parameters), and $\mathcal{L}(Y) = \prod_{t=1}^{r} p_I(I_t \mid Y_{0:t-1})$ for the integrand's likelihood factor, the target is

$$L_{\mathrm{SSI}} \;=\; \int \mathcal{L}(Y)\, \pi(Y)\, \mathrm{d}Y \;=\; \mathbb{E}_{Y\sim\pi}\big[\mathcal{L}(Y)\big]. \tag{45}$$

The unnormalised posterior (the joint integrand) is $\tilde{p}(Y) = \mathcal{L}(Y)\,\pi(Y)$, and the posterior density of the latents given the observations is $p_{\mathrm{post}}(Y) = \tilde{p}(Y)/L_{\mathrm{SSI}}$. The MCMC sampler of § 2.1.2 produces draws $Y^{p,(j)} \sim p_{\mathrm{post}}$ for $j = 1, \dots, N_1$.

5.1 Naive Monte Carlo

Equation (45) is itself an expectation under the prior π, so the simplest estimator just averages the conditional likelihood at independent draws from π:

$$\widehat{L}_{\mathrm{SSI}}^{\mathrm{naive}} \;=\; \frac{1}{N}\sum_{n=1}^{N} \mathcal{L}\big(Y^{(n)}\big), \qquad Y^{(n)} \overset{\mathrm{iid}}{\sim} \pi. \tag{46}$$

No MCMC trace is needed: drawing $Y^{(n)}$ requires only independent Gamma(kI_t, k) samples for each $t \in \mathcal{T}$. In log space the implementation is

$$\log \widehat{L}_{\mathrm{SSI}}^{\mathrm{naive}} \;=\; \underset{n=1,\dots,N}{\mathrm{logsumexp}} \log \mathcal{L}(Y^{(n)}) \;-\; \log N.$$

The estimator is unbiased. Its relative variance is $\mathrm{Var}_\pi[\mathcal{L}]/(N L_{\mathrm{SSI}}^2)$, equivalently $\big(\chi^2(p_{\mathrm{post}} \,\|\, \pi) + 1\big)/N - 1/N$, where $\chi^2(p \,\|\, q) = \int (p/q - 1)^2 q$ is the χ^2 divergence. The χ^2 between prior and posterior is small whenever the data are weakly informative about the latents — and remarkably, in the SSI model this remains true even on moderately informative histories, because the prior variance I_t/k scales automatically with the cohort size I_t and so retains overlap with the posterior. The estimator becomes unreliable only when the data are very informative (e.g. long histories with substantial incidence at small k).

5.2 Importance sampling

Importance sampling generalises the naive estimator by sampling from a proposal g chosen to better cover the posterior:

$$\widehat{L}_{\mathrm{SSI}}^{\mathrm{IS}} \;=\; \frac{1}{N}\sum_{n=1}^{N} \frac{\tilde{p}\big(Y^{(n)}\big)}{g\big(Y^{(n)}\big)}, \qquad Y^{(n)} \overset{\mathrm{iid}}{\sim} g. \tag{47}$$

The estimator is unbiased for any proposal g with $g(Y) > 0$ wherever $\tilde{p}(Y) > 0$; its relative variance is $\chi^2(p_{\text{post}} \,\|\, g)/N$, minimised by taking g as close to p_{post} as possible.

We take

$$g(Y) \;=\; \prod_{t\in\mathcal{T}} \text{Gamma}(Y_t;\, \alpha_t, \beta_t), \tag{48}$$

i.e. a product of independent Gammas in the same family as the prior, with shape and rate parameters fitted by method of moments to the per-day marginals of the MCMC posterior trace $\{Y^{p,(j)}\}_{j=1}^{N_1}$:

$$\alpha_t \;=\; \frac{\bar{Y}_t^{\,2}}{S_t^2}, \qquad \beta_t \;=\; \frac{\bar{Y}_t}{S_t^2}, \qquad t \in \mathcal{T}, \tag{49}$$

where $\bar{Y}_t$ and S_t^2 are the sample mean and (unbiased) sample variance of $\{Y_t^{p,(j)}\}_j$. Matching the proposal family to the prior keeps importance weights bounded; matching its marginals to the MCMC posterior makes it close to p_{post} marginally.

Remark (Caveat: posterior correlations)**.** Equation (48) is a product proposal, while the joint posterior of $Y_{\mathcal{T}}$ may have non-trivial correlations arising from the shared force-of-infection terms. When such correlations are strong, $\chi^2(p_{\text{post}} \,\|\, g)$ does not shrink to zero in the large-data limit and the IS estimator's variance remains bounded below by that mismatch. Bridge sampling (§ 5.3) is more forgiving of proposal mismatch.

5.3 Bridge sampling

Bridge sampling uses draws from *both* the proposal g and the posterior p_{post}, weighted by a bridge function α. The starting identity (Meng & Wong, 1996) is the algebraic equality

$$L_{\text{SSI}} \;=\; \frac{\mathbb{E}_{Y\sim g}\big[\tilde{p}(Y)\,\alpha(Y)\big]}{\mathbb{E}_{Y\sim p_{\text{post}}}\big[g(Y)\,\alpha(Y)\big]}, \tag{50}$$

valid for any α such that $0 < \int g(Y)\tilde{p}(Y)\alpha(Y)\,\mathrm{d}Y < \infty$. The identity is verified by writing the numerator as $\int \tilde{p}\,g\,\alpha\,\mathrm{d}Y$ and the denominator as $L_{\text{SSI}}^{-1}\int \tilde{p}\,g\,\alpha\,\mathrm{d}Y$. Replacing the expectations by Monte Carlo averages over $\{Y^{q,(i)}\}_{i=1}^{N_2} \overset{\text{iid}}{\sim} g$ and $\{Y^{p,(j)}\}_{j=1}^{N_1} \sim p_{\text{post}}$ (the MCMC trace) gives the bridge-sampling estimator.

The asymptotic variance of $\widehat{L}_{\text{SSI}}$ depends on the choice of α. Minimising it (Meng & Wong, 1996) yields the optimal bridge

$$\alpha^*(Y) \;=\; \frac{C}{s_1\,g(Y) + s_2\,\tilde{p}(Y)/L_{\text{SSI}}}, \tag{51}$$

where $s_1 = N_1/(N_1+N_2)$, $s_2 = N_2/(N_1+N_2)$, and C is any non-zero constant (any rescaling of α cancels between numerator and denominator of (50)). The optimal bridge depends on the unknown L_{SSI}; substituting (51) into (50) and replacing the expectations by Monte Carlo averages produces the fixed-point identity

$$L_{\text{SSI}} \;=\; \frac{\dfrac{1}{N_2}\displaystyle\sum_{i=1}^{N_2} \frac{\tilde{p}(Y^{q,(i)})}{s_1\,g(Y^{q,(i)}) + s_2\,\tilde{p}(Y^{q,(i)})/L_{\text{SSI}}}}{\dfrac{1}{N_1}\displaystyle\sum_{j=1}^{N_1} \frac{g(Y^{p,(j)})}{s_1\,g(Y^{p,(j)}) + s_2\,\tilde{p}(Y^{p,(j)})/L_{\text{SSI}}}}. \tag{52}$$

Solving (52) for L_{SSI} requires iterating the right-hand side from an initial guess. Bridge sampling has lower (or equal) asymptotic variance than importance sampling with the same proposal, and the variance bound is governed by a symmetrised divergence rather than $\chi^2(p_{\text{post}} \,\|\, g)$, so it is significantly more robust to proposal–posterior mismatch.

In practice we work entirely in log space for numerical stability. Define the log importance ratios at the two sample sets,

$$\ell_q^{(i)} \;=\; \log\tilde{p}(Y^{q,(i)}) - \log g(Y^{q,(i)}), \qquad \ell_p^{(j)} \;=\; \log\tilde{p}(Y^{p,(j)}) - \log g(Y^{p,(j)}).$$

Multiplying numerator and denominator of (52) by $1/g(Y^{q,(i)})$ and $1/g(Y^{p,(j)})$ respectively and taking logs gives the log-space iteration

$$\log L^{(t+1)} \;=\; \operatorname*{logsumexp}_{i=1,\dots,N_2}\Big[\,\ell_q^{(i)} - \mathrm{lse}\big(\log s_1,\, \log s_2 + \ell_q^{(i)} - \log L^{(t)}\big)\Big] - \log N_2 \\ -\; \operatorname*{logsumexp}_{j=1,\dots,N_1}\Big[-\mathrm{lse}\big(\log s_1,\, \log s_2 + \ell_p^{(j)} - \log L^{(t)}\big)\Big] + \log N_1, \quad (53)$$

where $\mathrm{lse}(a,b) = \log(e^a + e^b)$. We use the same Gamma proposal g as in § 5.2, initialise $\log L^{(0)}$ at the median of $\{\ell_q^{(i)}\}_i$, and iterate (53) until $\log L^{(t)}$ stops changing appreciably.

6 Uncertainty in the parameters R_0 and k

The preceding sections have treated the offspring-distribution parameters as known. In practice they are estimated from the data; we equip them with priors and ask for the resulting marginal model evidence and probability of a major outbreak. Write θ for the collection of parameters of model M ($\theta = (R_0, k)$ for SSE and SSI, $\theta = R_0$ for Poisson), and assume the prior factorises as a product of Gamma or LogNormal factors on the individual components; a fixed parameter corresponds to its prior factor being a point mass, which drops out of the integrals below without affecting the structure of any estimator.

Conditional on θ, each model has a per-parameter evidence $L_M(\theta) = \mathbb{P}_M(I_1, \dots, I_r \mid I_0, \theta)$ and PMO $\mathrm{PMO}_M(\theta) = 1 - q_r^M(\theta)$: the SSE evidence is the closed-form negative-binomial product (20), the Poisson evidence is the closed-form Poisson product (21), and the SSI evidence is the latent integral (23). The per-parameter PMOs are the parameter-known quantities of §§ 2.1.1–2.1.3.

6.1 Marginal evidence and model-averaged PMO

Define the prior-marginal evidence

$$\bar{L}_M \;=\; \int L_M(\theta)\, \pi(\theta)\, \mathrm{d}\theta, \tag{54}$$

and the corresponding marginal probability of a major outbreak,

$$\overline{\mathrm{PMO}}_M \;=\; \int \mathrm{PMO}_M(\theta)\, p(\theta \mid I_{1:r}, M)\, \mathrm{d}\theta \;=\; \frac{1}{\bar{L}_M} \int \mathrm{PMO}_M(\theta)\, L_M(\theta)\, \pi(\theta)\, \mathrm{d}\theta, \tag{55}$$

the posterior expectation of $\mathrm{PMO}_M(\theta)$ under the parameter posterior of model M. In the limit of a degenerate prior at θ_0, (54) reduces to $L_M(\theta_0)$ and (55) reduces to $\mathrm{PMO}_M(\theta_0)$, recovering the parameter-known quantities of the preceding sections.

The Bayesian model-averaging formulae (17) and (18) of § 3 carry over unchanged with $\bar{L}_M$ and $\overline{\mathrm{PMO}}_M$ in place of their parameter-known counterparts: the posterior model probabilities are $\pi_M^\star \propto \pi_M\, \bar{L}_M$ and the model-averaged probability of a major outbreak is $\overline{\mathrm{PMO}}_{\mathrm{avg}} = \sum_{M \in \mathcal{M}} \pi_M^\star\, \overline{\mathrm{PMO}}_M$. The two-model SSE/SSI case of (19) is the worked specialisation.

6.2 Estimators on the augmented latent vector

The three estimators of § 5 carry over to (54) after augmenting the latent vector with the uncertain parameters. Write $\theta = (\theta_{\mathrm{u}}, \theta_{\mathrm{f}})$ where θ_{u} collects the components of θ carrying a non-degenerate prior factor and θ_{f} collects the remaining (fixed) components. The augmented latent vector is

$$z \;=\; (\theta_{\mathrm{u}},\, Y_{\mathcal{T}}),$$

where $Y_{\mathcal{T}}$ is the SSI latent infectivities on the active days $\mathcal{T} = \{t \in \{0, \dots, r-1\} : I_t > 0\}$; for the SSE and Poisson models this block is empty and $z = \theta_{\mathrm{u}}$. The unnormalised target on z is

$$\tilde{p}(z) \;=\; \mathcal{L}\big(z;\, \theta_{\mathrm{f}}\big)\, \pi_Y\big(Y_{\mathcal{T}} \mid k\big)\, \pi(\theta_{\mathrm{u}}), \tag{56}$$

where $\mathcal{L}(z; \theta_\mathrm{f}) = p(I_{1:r} \mid z, \theta_\mathrm{f})$ is the observation likelihood — the SSE NB product (20) for SSE, the Poisson product (21) for Poisson, and the latent-conditional Poisson product $\prod_{t=1}^{r} p_I(I_t \mid Y_{0:t-1})$ of (22) for SSI — and π_Y is the SSI latent prior of § 5 (absent in the SSE and Poisson cases). By construction

$$\bar{L}_M \;=\; \int \tilde{p}(z)\,\mathrm{d}z, \tag{57}$$

so $\bar{L}_M$ is the normalising constant of the augmented target and $p_\mathrm{post}(z) = \tilde{p}(z)/\bar{L}_M$ is the augmented posterior. When θ_u is empty, (56) collapses to the latent-only target $\tilde{p}(Y)$ of (45) and the development below reduces to that of § 5.

Naive Monte Carlo. Draw $\theta_\mathrm{u}^{(n)} \sim \pi$ and, for the SSI model, $Y_\mathcal{T}^{(n)} \mid k^{(n)} \sim \pi_Y(\cdot \mid k^{(n)})$, jointly forming $z^{(n)}$. The estimator

$$\widehat{\bar{L}}_M^{\,\mathrm{naive}} \;=\; \frac{1}{N}\sum_{n=1}^{N} \mathcal{L}\big(z^{(n)};\, \theta_\mathrm{f}\big) \tag{58}$$

is unbiased. For the SSE and Poisson models the latent block is empty and $\mathcal{L}(z; \theta_\mathrm{f})$ is the closed-form $L_M(\theta^{(n)})$ of (20) or (21); no MCMC is required. Naive MC inherits the variance pathology of § 5.1: it is reliable when the prior and the augmented posterior overlap broadly, and degrades on informative histories where the posterior concentrates in a region of low prior mass.

Importance sampling. Run HMC against $\tilde{p}$ of (56) to obtain posterior draws of the augmented vector z. Fit an independent-Gamma proposal

$$g(z) \;=\; \prod_j \mathrm{Gamma}\big(z_j;\, \alpha_j, \beta_j\big) \tag{59}$$

by method of moments (49) to each marginal of the trace (the index j now ranges over the components of θ_u as well as the active days $\mathcal{T}$). The importance-sampling estimator

$$\widehat{\bar{L}}_M^{\,\mathrm{IS}} \;=\; \frac{1}{N}\sum_{n=1}^{N} \frac{\tilde{p}\big(z^{(n)}\big)}{g\big(z^{(n)}\big)}, \qquad z^{(n)} \overset{\mathrm{iid}}{\sim} g, \tag{60}$$

is the literal extension of (47); the variance discussion of § 5.2 applies verbatim with $\chi^2(p_\mathrm{post} \,\|\, g)$ now measured on the augmented space. The Gamma proposal again matches the support of every component of z: the latent infectivities are Gamma a priori, and both Gamma and LogNormal parameter priors are supported on the positive reals.

Bridge sampling. With $\{z^{p,(j)}\}_{j=1}^{N_1}$ from the HMC trace and $\{z^{q,(i)}\}_{i=1}^{N_2}$ from the proposal (59), define the augmented log importance ratios

$$\ell_q^{(i)} \;=\; \log \tilde{p}\big(z^{q,(i)}\big) - \log g\big(z^{q,(i)}\big), \qquad \ell_p^{(j)} \;=\; \log \tilde{p}\big(z^{p,(j)}\big) - \log g\big(z^{p,(j)}\big).$$

The Meng–Wong identity (50) and its optimal bridge α^* are unchanged in form (with $\bar{L}_M$ replacing L_SSI and the augmented variables in place of Y), and the log-space fixed-point iteration (53) applies verbatim.

6.3 Simulation-based rejection scheme

The rejection sampler of § 3.2 extends without modification: for each simulated trajectory under model M, draw a fresh $\theta^{(i)} \sim \pi$, simulate the SSE / SSI / Poisson generative process forward with parameters $\theta^{(i)}$, and retain only the trajectories whose first r cohorts match the observed $I_{1:r}$. The acceptance probability of a single M-trajectory is $\int p(I_{1:r} \mid \theta, M)\,\pi(\theta)\,\mathrm{d}\theta = \bar{L}_M$, so the empirical fraction of accepted trajectories that go on to a major outbreak is consistent for $\overline{\mathrm{PMO}}_M$ of (55). Pooling trajectories across the candidate set $\mathcal{M}$ with per-batch model split Multinomial$(b, (\pi_M)_{M\in\mathcal{M}})$ realises the posterior model partition with expected fractions $\pi_M\,\bar{L}_M$, so the composition of the accepted pool delivers the N-model average $\overline{\mathrm{PMO}}_\mathrm{avg}$ of (55) together with (17)–(18), exactly as the two-model SSE/SSI case of § 3.2.

7 Symptom-onset data and delayed transmission

The models of §1 treat the observed history $I_0, I_1, \ldots$ as a record of *infection* times. In practice surveillance data record *symptom onset* times, and transmission is naturally referenced to a case's own onset rather than to the (unobserved) moment of infection. For a disease with negligible pre-symptomatic transmission — Ebola virus disease is our motivating example — we can build a renewal process directly on the onset times, driven by the distribution of the *time from symptom onset to transmission* (TOST), and connect infections to their eventual onsets through an incubation-period distribution.

Let $D_t \geq 0$ denote the number of individuals whose symptoms begin (onset) on day t; this is the observed quantity. We introduce two discrete distributions on the non-negative integers:

- the *incubation-period* weights ι_a for $a = 1, 2, \ldots$, with $\sum_{a\geq 1} \iota_a = 1$, giving the probability that an infected individual develops symptoms a days after infection;
- the *TOST* weights τ_s for $s = 0, 1, 2, \ldots$, with $\sum_{s\geq 0} \tau_s = 1$, giving the probability that a transmission event occurs s days after the infector's own onset.

Presymptomatic transmission would place mass on $s < 0$ and is assumed negligible here. The incubation distribution is indexed from $a = 1$ and the TOST from $s = 0$; the significance of this asymmetry is discussed below.

7.1 Onset-anchored SSE and SSI models

Write J_t for the number of *new infections* generated on day t. The force of infection is now the TOST-weighted sum of past *onset* cohorts. In the onset-anchored SSE model,

$$\lambda_t = \sum_{s=0}^{\infty} \tau_s \, D_{t-s}, \qquad J_t \mid \lambda_t \;\sim\; \mathrm{NB}(\text{mean} = R_0\lambda_t,\ \text{disp} = k\lambda_t)\,. \tag{61}$$

In the onset-anchored SSI model each onset cohort carries an aggregate latent infectivity, and infections arise as a Poisson process driven by past infectivity,

$$Y_t \mid D_t \;\sim\; \mathrm{Gamma}(\text{shape} = kD_t,\ \text{rate} = k)\,, \qquad \lambda_t = \sum_{s=0}^{\infty} \tau_s \, Y_{t-s}, \qquad J_t \mid \lambda_t \;\sim\; \mathrm{Poisson}(R_0\lambda_t)\,, \tag{62}$$

exactly as in §1.2 but with the serial-interval weights w_s replaced by the TOST weights τ_s and the driving cohorts taken to be onsets. As $k \to \infty$ both reduce to a common onset-anchored Poisson model with $J_t \sim \mathrm{Poisson}(R_0\lambda_t)$ and $\lambda_t = \sum_{s\geq 0} \tau_s D_{t-s}$.

Infections are mapped forward to onsets through the incubation period: each of the J_t infections on day t independently draws an incubation period $A \sim (\iota_a)_{a\geq 1}$ and contributes to the onset cohort on day $t + A$. Equivalently,

$$D_t = \sum_{a=1}^{\infty} J_{t-a}^{(a)}, \qquad \big(J_u^{(a)}\big)_{a\geq 1} \,\big|\, J_u \;\sim\; \mathrm{Multinomial}\big(J_u,\, (\iota_a)_{a\geq 1}\big), \tag{63}$$

where $J_u^{(a)}$ is the number of infections on day u whose symptoms begin a days later. The generative process alternates $D_t \to \{Y_t\} \to J_t \to D_{>t}$, seeded by a single index case with onset on day 0 ($D_0 = 1$).

Indexing convention and causal ordering. Because the incubation weights are supported on $a \geq 1$, an infection on day t cannot produce an onset before day $t + 1$. Consequently the onset cohort D_t — and, in the SSI model, its infectivity Y_t — is completely determined by infections on days strictly before t, so it is fully realised before day t's own transmission is evaluated in (61)–(62). This well-orders the recursion. The TOST weights, by contrast, are supported on $s \geq 0$, so a case may transmit on its own onset day. At most one of the two distributions may place mass at lag 0: if both did, a chain of same-day infection, onset and transmission could occur within a single day and the recursion would no longer be well-defined.

Relationship to the serial interval. Consider an infector with onset on day u. It transmits on day $u+S$ with $S \sim \tau$; the infectee, infected on day $u+S$, has onset on day $u+S+A'$ with $A' \sim \iota$. The onset-to-onset serial interval is therefore $S+A'$, distributed as the convolution $\tau * \iota$ with mean $\mathbb{E}[\tau]+\mathbb{E}[\iota]$. Choosing the TOST and incubation means so that they sum to the serial-interval mean used in §1 yields an onset-anchored model that is roughly comparable to the infection-anchored models, while making explicit the intermediate onset-to-transmission and infection-to-onset delays.

7.2 Probability of a major outbreak

Given an observed onset history $D_0, \ldots, D_r$ (with $D_0 \geq 1$), we again ask for the probability of a major outbreak. Unlike the infection-anchored SSE model of §2.1.1, the observed onsets do not determine the state needed to project the epidemic forward: at the observation horizon there is a pipeline of infected-but-not-yet-symptomatic individuals, whose number and future onset days are latent, together with (in the SSI model) the latent infectivities. We therefore estimate the probability of a major outbreak by the simulation-based rejection scheme of §3.2, applied to the onset process: simulate onset trajectories forward from a single index onset on day 0, retain those whose onset cohorts on days $1, \ldots, r$ match the observed $D_1, \ldots, D_r$, and continue each retained trajectory until it either reaches the major-outbreak threshold (a single-day onset count at or above the threshold) or goes extinct. Extinction is declared once no infected individuals remain in the incubation pipeline *and* no onsets have occurred within the TOST support of recent days, so that no further transmission is possible; tracking the pipeline (infections generated minus onsets realised) is necessary because a recent run of zero onsets alone does not preclude onsets already scheduled from earlier infections.

This scheme applies unchanged to the onset-anchored SSE, SSI and Poisson models. Closed-form expressions for the residual extinction probability, analogous to those of §2.1, are likely tractable — the SSE and Poisson offspring still decompose independently, and the SSI infectivities admit the same conditioning arguments — but are not pursued here; the simulation-based estimator suffices for the exploratory comparisons of interest.

Real-time monitoring. Independent rejection sampling becomes inefficient when the probability of a major outbreak is required *sequentially*, at each week of an unfolding outbreak, since matching the full observed onset history to week t has an acceptance probability that decays rapidly in t. A bootstrap particle filter is far more efficient: a population of trajectories is advanced one week at a time, and at each week the particles whose newly realised onset cohort matches the observation are retained and resampled with replacement, so the hard-won early-history match is reused rather than re-simulated. Because the incubation delay is at least one time unit, the week-t onset cohort is already determined by the state carried in each particle, so conditioning on it requires no fresh simulation; the retained particles then represent the distribution of the latent state given the onsets to week t, and simulating each forward freely to resolution yields the probability of a major outbreak conditional on the history observed so far.

Reporting delays (future extension). A natural refinement retains this onset process model but adds a reporting delay: at the time of analysis some symptom onsets have not yet been reported, so the observed history is a right-truncated and thinned version of the true onset history. The same simulation-based framework accommodates this by matching only the reported subset of onsets (or by marginalising over unreported recent onsets) when accepting trajectories. We note it here as a direction for future work and do not develop it further.